

Exercise sheet 1

Topics in Topology, MTH602

1. What is $\pi_1(X \times Y, (x_0, y_0))$ in terms of $\pi_1(X, x_0)$, and $\pi_1(Y, y_0)$?
2. Show that f_* is injective where $f : S^1 \rightarrow S^1 \times S^1$ defined as $f(t) = (s_0, t)$.